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Securing Technological Leadership? The Cost of Export Controls on Firms

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1 Big picture

- **Question.** What are the firm-level and supply-chain costs of U.S. export controls that restrict domestic firms from selling advanced technologies to selected Chinese firms?
- **Setting.** The paper studies export controls administered by the Bureau of Industry and Security (BIS), especially additions of Chinese firms to the Entity List, Military End User List, and Unverified List. These lists restrict or complicate U.S. firms' ability to export goods, software, and technology to targeted Chinese entities.
- **Policy motivation.** Export controls are intended to safeguard U.S. technological leadership, especially in strategic sectors such as semiconductors, wireless communications, and advanced computing. The central policy trade-off is whether national-security benefits come at high costs to the domestic firms producing the protected technologies.
- **Main empirical idea.** The paper uses hand-collected BIS list events matched to global firm-to-firm supply-chain links. It compares U.S. suppliers connected to Chinese targets with similar U.S. firms exporting to Chinese customers that are not targeted.
- **Main supply-chain result.** Affected U.S. suppliers do decouple from targeted Chinese customers. But the decoupling is broader than intended: they also terminate relationships with other Chinese customers and form fewer new Chinese customer relationships.
- **Main reallocation result.** Affected U.S. suppliers do not quickly replace lost Chinese customers with new domestic customers, customers in allied Asian countries, or customers in Europe. The short-run customer base becomes smaller rather than reconfigured.
- **Main stock-market result.** Affected U.S. suppliers experience negative abnormal returns around export-control announcements. The paper reports an average cumulative abnormal return of about **-3.6%** over the event window, implying roughly **\$158 billion** in aggregate U.S. supplier market-capitalization losses.
- **Main real-effect result.** Export controls reduce affected suppliers' cash flow, revenue, EBIT, and employment. The key estimates imply revenue falls by about **8.9%**, employment by about **7.3%**, and EBIT by about **28%** of the treated-firm average.
- **Main financing result.** Affected suppliers face tighter bank credit conditions: lower total credit and term loans, higher spreads, and shorter maturities.
- **Chinese-side result.** Targeted Chinese firms respond more actively. They terminate U.S. supplier links, create new domestic Chinese supplier links, increase their Chinese supplier share, and reduce their U.S. supplier share.
- **Circumvention result.** Non-U.S. suppliers connected to Chinese targets, especially suppliers located in allied regions, experience revenue gains. This suggests that Chinese targets may partly substitute toward non-U.S. suppliers outside the direct reach of U.S. export controls.
- **Overall takeaway.** Export controls achieve immediate decoupling, but they impose large collateral costs on the domestic firms that develop the technologies the policy aims to protect. Chinese targets also suffer, but the aggregate stock-market losses of U.S. suppliers are larger.

2 Data and measurement (institutional context and key objects)

2.1 Policy background: BIS export-control lists

- The relevant regulator is the **Bureau of Industry and Security** under the U.S. Department of Commerce.
- The central legal framework is the Export Administration Regulations.
- The paper focuses on three BIS lists:
 - **Entity List:** foreign persons or entities subject to license requirements because they are viewed as contrary to U.S. national-security or foreign-policy interests;
 - **Military End User List:** entities posing military end-use or military end-user risks;
 - **Unverified List:** entities for which BIS cannot verify end use or end user.
- Inclusion in the Entity List or Military End User List is more restrictive than inclusion in the Unverified List.

Interpretation. The policy is not a tariff on imports. It is a restriction on U.S. firms' ability to supply particular foreign customers with strategic technology. The direct intended channel is forced seller-customer separation.

2.2 Hand-collected export-control data

- The authors hand collect additions and removals of Chinese entities from BIS lists using Federal Register and Code of Federal Regulations sources.
- They collect aliases, announcement dates, effective dates, and physical addresses.
- Excluding aliases, the sample includes **732 unique Chinese entities: 470 corporations and 262 universities or institutions**.
- By list type, there are **425** Entity List entities, **58** Military End User List entities, and **253** Unverified List entities. These counts overlap because some entities appear on multiple lists over time.

Interpretation. The treatment event is not self-reported by firms. It comes from regulatory actions that identify specific Chinese entities and dates.

2.3 Supply-chain data and affected suppliers

- Supply-chain links come from **FactSet Revere**, which records supplier-customer relationships, relation start dates, and relation end dates.
- The sample covers supply-chain relations through **2023:Q3**.

- The authors match **90 Chinese target entities** to supply-chain data.
- These targets are connected to **351 affected suppliers**; **176** of these have a supply-chain relation that overlaps the export-control event date.
- Fig. 1 shows that affected suppliers are concentrated in technology-related industries, especially electronic equipment, industrial machinery, business services, instruments, and transportation equipment.

Interpretation. The data identify the exact domestic firms whose existing customers are hit by export controls. This lets the paper study both intended decoupling and unintended business losses.

2.4 Financial and real-outcome data

- Daily stock-price data come from **CRSP**; balance-sheet data come from **Compustat North America**.
- The daily stock-price event sample contains **250 events** involving **156 affected suppliers** from **2010–2022**.
- The annual balance-sheet panel runs from **2007–2022** and contains **655 firms**, of which **126** are affected suppliers.
- All firms in the balance-sheet panel export to China, and firms with less than **\$5 million** in total assets are removed.

Interpretation. The paper compares affected exporters to China with other exporters to China, rather than comparing China-exposed technology firms with unrelated firms.

2.5 Bank-lending data

- Loan-level data come from the Federal Reserve’s confidential **Y-14Q** corporate-loan schedule.
- The bank-lending sample contains **331 firms**, including **71 affected firms**, borrowing from **38 banks** from **2012:Q3–2023:Q3**.
- The loan outcomes include total credit commitments, term loans, credit lines, utilization, spreads, and maturities.

Interpretation. The lending data allow the paper to ask whether export controls affect not only firms’ product-market relationships but also their access to bank finance.

2.6 Chinese-target and international-supplier data

- Chinese stock-price data come from Refinitiv; China-specific factor returns follow Liu et al. (2019).
- For international suppliers, the paper uses Capital IQ data on EBIT and revenue.

- The international supplier sample includes **6,068 suppliers** of Chinese firms, of which **525** are connected to Chinese firms targeted by export controls.

Interpretation. The Chinese-side data let the paper study whether Chinese targets simply lose U.S. suppliers or actively rebuild supply chains through domestic and non-U.S. alternatives.

2.7 Main variables

- **Affected:** equals one for a U.S. supplier whose Chinese customer is included in a BIS list.
- **Post:** equals one after the Chinese customer is included in the BIS list.
- **Targeted:** equals one for a Chinese firm directly included in a BIS list.
- **Supply-chain outcomes:** terminated Chinese customers, terminated Chinese customers excluding the directly targeted customer, new Chinese customers, total customers, domestic customers, and customer shares by region.
- **Financial and real outcomes:** cumulative abnormal returns, cash flow, revenue, EBIT, capital expenditures, employment, and bank-lending terms.

Interpretation. The paper separates the direct intended effect, termination of target links, from broader spillovers, loss of non-target Chinese customers, inability to replace customers, real losses, and financing deterioration.

3 Models and empirical specifications (all equations + interpretation)

3.1 Stacked difference-in-differences design

The main estimating equation is a stacked regression:

$$y_{ict} = \beta(Affected_{ic} \times Post_{ct}) + \mu_{ic} + \mu_{ckt} + \varepsilon_{ict}. \quad (1)$$

Here:

- i indexes a U.S. supplier;
- c indexes a cohort, meaning a specific round of export controls;
- t indexes time;
- k indexes industry and size controls;
- μ_{ic} are cohort-firm fixed effects;
- μ_{ckt} are cohort-specific industry-size-time fixed effects.

Interpretation. The coefficient β compares affected suppliers with control suppliers in the same cohort and similar industry-size-time environment before versus after the export-control event.

3.2 Why the paper uses stacked cohorts

- Export controls occur in many different years, so the treatment is staggered.
- A standard two-way fixed effects design can be biased with staggered treatment timing.
- The paper therefore stacks event-specific cohorts in windows around each event.
- The control group includes firms that are never treated or not yet treated.

Interpretation. The stacked design avoids using already-treated firms as inappropriate controls for newly treated firms.

3.3 Count outcomes and PPML

Many supply-chain variables are counts or count-like variables, such as the number of terminated customers or new supplier relations. The paper estimates these using Poisson pseudo-maximum likelihood (PPML):

$$\mathbb{E}[y_{ict} | X_{ict}] = \exp(\beta Affected_{ic} \times Post_{ct} + \mu_{ic} + \mu_{ckt}). \quad (2)$$

Interpretation. With PPML, a coefficient can be translated into a proportional effect using $\exp(\beta) - 1$. For example, a coefficient of 0.560 implies approximately $\exp(0.560) - 1 \approx 75\%$ more terminations.

3.4 Dynamic event-study specification

To examine pre-trends and timing, the paper estimates dynamic versions of the stacked regression:

$$y_{ict} = \sum_{\tau \neq -1} \beta_{\tau} (Affected_{ic} \times \mathbf{1}\{t - T_c = \tau\}) + \mu_{ic} + \mu_{ckt} + \varepsilon_{ict}. \quad (3)$$

The omitted period is typically the year before treatment, $\tau = -1$.

Interpretation. The dynamic coefficients show whether affected and control firms were already moving differently before export controls. The figures show little evidence that the main patterns are driven by pre-trends.

3.5 Stock-market event study

The stock-market analysis estimates abnormal returns using factor models. For U.S. suppliers, the benchmark factor model is:

$$r_{it} - r_{ft} = \alpha_i + \gamma_i' F_t + u_{it}, \quad (4)$$

where F_t is either the Fama–French 3-factor or 5-factor vector. Betas are estimated in a pre-event window, and cumulative abnormal returns are computed as:

$$CAR_i[a, b] = \sum_{t=a}^b [(r_{it} - r_{ft}) - \hat{\alpha}_i - \hat{\gamma}'_i F_t]. \quad (5)$$

The main event window is $[-10, 20]$ trading days around the BIS announcement.

Interpretation. The stock-market exercise asks whether investors immediately revalue U.S. suppliers when their Chinese customers are targeted by export controls.

3.6 Real effects on U.S. suppliers

The balance-sheet regressions use the same stacked design, with outcomes:

$$CashFlow_{ict}, \quad Revenues_{ict}, \quad EBIT_{ict}, \quad CapEx_{ict}, \quad Employees_{ict}. \quad (6)$$

Interpretation. These regressions test whether stock-market losses reflect real operating deterioration rather than only investor sentiment.

3.7 Bank-lending specification

The bank-lending analysis adds bank-time controls. A simplified specification is:

$$LoanOutcome_{ibct} = \beta(Affected_{ic} \times Post_{ct}) + \eta_{ict} + \lambda_{kct} + \eta_{bt} + \varepsilon_{ibct}, \quad (7)$$

where b indexes banks and η_{bt} are bank-quarter fixed effects.

Interpretation. Bank-quarter fixed effects compare affected and control firms borrowing from the same bank in the same quarter, helping separate borrower deterioration from bank-wide credit-supply shocks.

3.8 Chinese-target regressions

For Chinese firms, the main regressor becomes $Targeted \times Post$:

$$y_{jct} = \beta(Targeted_{jc} \times Post_{ct}) + \mu_{jc} + \mu_{ct} + \varepsilon_{jct}, \quad (8)$$

where j indexes Chinese firms.

Interpretation. This asks whether the Chinese target terminates U.S. supplier links and whether it replaces them with Chinese suppliers or non-U.S. suppliers.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the U.S.-supplier estimates

- Treatment is triggered by the BIS listing of a supplier's Chinese customer.
- The comparison group is other U.S. firms that also export to China, but whose Chinese customers are not targeted at that time.
- Cohort-firm fixed effects absorb time-invariant differences between treated and control firms.
- Cohort-industry-size-time fixed effects absorb shocks common to similar firms in the same period.

Interpretation. Identification comes from within-cohort changes in outcomes for affected suppliers relative to similar China exporters that are not directly exposed to the same listing event.

4.2 Why the decoupling result is especially clean

- Export controls directly restrict transactions with the targeted Chinese entity.
- The most immediate outcome, relationship termination, is the intended policy margin.
- The paper then checks whether effects extend beyond the direct target to non-target Chinese customers and new Chinese relationships.

Interpretation. Termination of the targeted relationship confirms compliance. Termination of non-target Chinese customers and reduced formation of new Chinese links reveal broader supply-chain disruption.

4.3 Addressing pre-trends

- Dynamic estimates in Fig. 2 show little evidence of differential pre-trends in U.S.-supplier decoupling outcomes.
- Fig. 4 similarly shows that Chinese supply-chain reconfiguration is not driven by pre-existing dynamics.
- Stock-market event studies show no strong abnormal returns immediately before the event but negative returns after announcement.

Interpretation. The timing of the outcomes is consistent with export-control announcements causing the disruptions rather than merely revealing trends already underway.

4.4 Alternative explanations

- **General China risk.** The control firms also export to China, so the comparison is not simply China-exposed versus non-China-exposed firms.
- **Industry shocks.** Industry-size-year fixed effects absorb shocks to similar firms in the same sector and size class.
- **Firm quality.** Cohort-firm fixed effects absorb persistent firm differences.
- **Bank supply shocks.** In lending regressions, bank-quarter fixed effects absorb changes in a bank's general lending stance.

Interpretation. The design cannot fully remove every concern, but it targets the most obvious confounds: China exposure, sector shocks, firm fixed differences, and bank-wide lending shocks.

4.5 What the paper can and cannot say

- The paper provides strong evidence on **costs to firms** and **supply-chain reconfiguration**.
- It does not directly measure classified national-security benefits or long-run strategic benefits.
- It shows that Chinese targets adjust supply chains, but it cannot fully measure whether export controls slow Chinese innovation or military capability in the long run.
- It suggests that export controls may require complementary policies that support domestic demand for the restricted technologies.

Interpretation. The paper is best read as a cost-side analysis of export controls, not a complete welfare analysis of national-security policy.

5 Tables and figures

5.1 Figure 1: affected U.S. suppliers

Read: The number of affected U.S. suppliers rises sharply after major BIS actions involving firms such as ZTE, Huawei, SMIC, and Yangtze. The most affected industries are technology-intensive, especially electronic equipment and industrial machinery.

Economic message. The policy shock is concentrated exactly where technological leadership matters most: advanced manufacturing, electronics, business services, and related high-tech supply chains.

5.2 Table 1: supply-chain summary statistics

Read: Treated suppliers have more Chinese customer terminations and more total customers than controls. They also have a higher Chinese-customer share before treatment, while controls have a

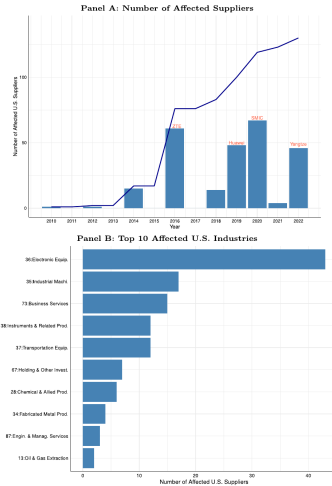


Fig. 1. Number of affected U.S. suppliers. Panel A displays the number of affected U.S. suppliers over time as the BIS includes Chinese customers in the BIS List. The histogram shows the number of affected U.S. suppliers in a specific year. The blue line represents the cumulative number of affected U.S. suppliers over time. Symbolic Chinese firms that are included in the BIS List are highlighted with orange text. Panel B displays the top 10 most affected industries based on the total number of affected U.S. suppliers in each industry. The industry classification is based on the 2-digit SIC code.

Table 1

Summary statistics—Supply chain reconfigurations. This table presents summary statistics for firms' supply chain relationships based on their treatment status (treated if they supply to Chinese entities in the BIS lists; control if they export to Chinese entities not in the BIS lists). Termination Chinese Cust is the total number of terminated relations with Chinese customers. Termination Chinese Cust (excl. targeted) is the total number of terminated relations with Chinese customers, excluding those targeted by the BIS lists. New Relations Chinese Cust is the number of new Chinese customers. Total Cust is the total number of customers. Domestic Cust is the number of domestic (U.S.) customers. Domestic Share is the ratio of the total number of domestic U.S. customers to the contemporaneous number of total customers. China share is the ratio of the total number of Chinese customers to the contemporaneous number of total customers. Asia share is the ratio of the total number of customers from Asian countries other than China to the contemporaneous number of total customers. Asia Friend Share is the ratio of the total number of customers from South Korea, Japan, Australia, and Taiwan to the contemporaneous number of total customers. EU share is the ratio of the total number of customers from European Union countries to the contemporaneous number of total customers. SD refers to standard deviation, Obs refers to the number of observations, and p(25), p(50), and p(75) refer to the 25th, 50th, and 75th percentiles, respectively.

	Mean	SD	Obs	p(25)	p(50)	p(75)
Termination Chinese Cust	0.209	0.738	5,145	0	0	0
Treated	0.556	1.391	744	0	0	1
Control	0.150	0.534	4,401	0	0	0
Termination Chinese Cust (excl.targeted)	0.197	0.706	5,145	0	0	0
Treated	0.476	1.293	744	0	0	0
Control	0.150	0.534	4,401	0	0	0
New Relations Chinese Cust	0.451	1.313	5,145	0	0	0
Treated	1.172	2.454	744	0	0	1.2
Control	0.329	0.947	4,401	0	0	0
Total Cust	33.464	61.891	5,145	8	19	39
Treated	61.801	120.421	744	17	34	59
Control	28.674	43.252	4,401	7	18	37
Domestic Cust	15.534	25.072	5,145	3	9	19
Treated	24.535	44.189	744	6	13	23
Control	14.013	19.727	4,401	3	8	18
Domestic Share	0.498	0.246	5,050	0.333	0.500	0.667
Treated	0.408	0.175	740	0.303	0.399	0.500
Control	0.513	0.253	4,310	0.333	0.500	0.664
China Share	0.064	0.134	5,050	0.000	0.015	0.071
Treated	0.093	0.110	740	0.027	0.064	0.117
Control	0.059	0.137	4,310	0.000	0.000	0.059
Asia Share	0.180	0.184	5,050	0.024	0.143	0.267
Treated	0.232	0.170	740	0.114	0.217	0.321
Control	0.171	0.184	4,310	0.000	0.125	0.250
Asia Friend Share	0.149	0.173	5,050	0.000	0.100	0.218
Treated	0.201	0.168	740	0.074	0.107	0.224
Control	0.140	0.173	4,310	0.000	0.088	0.200
EU Share	0.130	0.127	5,050	0.000	0.111	0.192
Treated	0.136	0.099	740	0.062	0.130	0.198
Control	0.129	0.131	4,310	0.000	0.106	0.190

higher domestic-customer share.

Economic message. Affected suppliers are not random U.S. firms. They are globally connected exporters with meaningful Chinese-customer exposure, which makes the policy shock economically important.

5.3 Table 2: financial and real summary statistics

Read: Treated suppliers are generally larger and more profitable than controls. The CAR summary statistics show negative post-event abnormal returns for affected U.S. suppliers.

Economic message. Export controls hit large, productive firms rather than marginal firms. Losses therefore matter for the very industrial base the policy is designed to protect.

5.4 Table 3: balance-sheet comparability within size quartiles

Read: After splitting by industry-year-size quartiles, treated and control firms look more comparable, especially outside the smallest size quartile.

Economic message. Size and industry controls are central to the design because affected suppliers tend to be larger and more technologically intensive in the raw data.

Table 2

Summary statistics—Financial and real collateral damage. This table presents summary statistics for firms' balance sheet characteristics based on their treatment status (treated if they supply to Chinese entities in the BIS lists; control if they export to Chinese entities not in the BIS lists) and for the cumulative abnormal returns of Treated suppliers before and after the announcement of export controls. SD refers to standard deviation, Obs refers to the number of observations, and p(25), p(50), and p(75) refer to the 25th, 50th, and 75th percentiles, respectively. Cash Flow equals operating income before depreciation minus interest and taxes, divided by lagged assets. Revenues is the logarithm of the total revenues (in millions). Sale is the logarithm of the total sales (in millions), ROA is return on assets, CapEx is capital expenditures divided by lagged assets, Income equals operating income before depreciation divided by lagged assets, EBIT is earnings before interest and taxes divided by lagged assets, interest is interest expenses divided by lagged assets, and Employees is the logarithm of the total number of employees. 3-factor CAR is cumulative abnormal return using the Fama-French 3-factor model (Fama and French, 1993). 5-factor CAR is cumulative abnormal return using the Fama-French 5-factor model (Fama and French, 2015). The two event windows $[-10, -1]$ and $[0, 20]$ represent trading days relative to the event date.

	Mean	SD	Obs	p(25)	p(50)	p(75)
Balance Sheet Characteristics						
Assets, \$m	11,946	48,333	5,119	222	1,019	4,624
Treated	14,891	41,328	741	442	1,854	7,588
Control	11,447	49,407	4,378	202	894	4,188
Cash Flow	0.015	0.279	5,092	-0.008	0.074	0.123
Treated	0.084	0.129	738	0.052	0.098	0.135
Control	0.003	0.295	4,354	-0.028	0.068	0.119
Revenues	6.533	2.221	5,082	5.116	6.699	8.114
Treated	7.119	2.019	740	5.831	7.155	8.589
Control	6.434	2.239	4,342	4.989	6.618	8.050
Sales	6.523	2.218	5,059	5.112	6.696	8.098
Treated	7.116	2.019	738	5.829	7.148	8.591
Control	6.422	2.234	4,321	4.976	6.615	8.034
ROA	-0.041	0.274	5,118	-0.075	0.027	0.081
Treated	0.031	0.141	741	0.002	0.048	0.092
Control	-0.053	0.288	4,377	-0.097	0.023	0.078
CapEx	0.034	0.042	5,089	0.012	0.042	0.041
Treated	0.038	0.056	738	0.013	0.024	0.041
Control	0.034	0.039	4,351	0.012	0.042	0.042
Income	0.036	0.263	5,092	0.006	0.098	0.154
Treated	0.107	0.131	738	0.074	0.122	0.165
Control	0.024	0.277	4,354	-0.015	0.092	0.151
EBIT	-0.001	0.272	5,095	-0.039	0.058	0.114
Treated	0.064	0.136	739	0.032	0.080	0.123
Control	-0.012	0.288	4,356	-0.057	0.052	0.111
Interest	0.014	0.028	4,602	0.002	0.008	0.017
Treated	0.010	0.011	679	0.004	0.008	0.013
Control	0.015	0.029	3,923	0.001	0.008	0.018
Employees	7.727	2.055	5,068	6.284	7.864	9.179
Treated	8.281	2.040	737	6.939	8.497	9.852
Control	7.633	2.043	4,331	6.210	7.722	9.083
Cumulative Abnormal Returns						
3-factor CAR						
[-10, -1]	-0.011	0.062	250	-0.053	-0.009	0.024
[0, 20]	-0.025	0.103	250	-0.081	-0.029	0.024
5-factor CAR						
[-10, -1]	-0.007	0.085	250	-0.047	-0.007	0.027
[0, 20]	-0.027	0.110	250	-0.086	-0.025	0.023

5.5 Table 4: decoupling from China

Read: Affected suppliers terminate more relationships with Chinese customers. The coefficients are positive for total Chinese-customer terminations and still positive when the directly targeted customer is excluded. The coefficients on new Chinese-customer relations are negative.

Economic message. Export controls do more than sever the legally restricted link. They trigger broader China decoupling and reduce the formation of new Chinese business relationships.

Table 4

Decoupling from China. This table presents the Poisson Pseudo Maximum Likelihood (PPML) regression results of the effect of export controls on supply chain configurations. Termination Chinese Cust is the total number of terminated relations with Chinese customers. Termination Chinese Cust (excl. targeted) is the total number of terminated relations with Chinese customers, excluding those targeted by the BIS lists. New Relations Chinese Cust is the number of new Chinese customers. Affected equals one for firms that within the previous year had a customer included in the BIS lists (Entity List, UFL, and MEU list) and Post equals one after the inclusion of such customer in the BIS lists. For each cohort, the control group includes never treated and not yet treated firms. SIC refers to the 2-digit standard industrial classification (SIC) code. Size refers to the industry-specific size quartile of each firm. Custom refers to the classification of each firm into quartiles based on treatment firms' lagged total number of customers. We require all firms to be exporting to China in the pre-treatment period. We double cluster the standard errors at the firm and year level. *** p<0.01, ** p<0.05, * p<0.1.

Dependent variables:	Termination Chinese Cust			Termination Chinese Cust (excl. targeted)			New Relations Chinese Cust		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Affected*Post	0.572*** (0.209)	0.590*** (0.234)	0.702*** (0.261)	0.376* (0.223)	0.414* (0.242)	0.560*** (0.262)	-0.483*** (0.139)	-0.523*** (0.153)	-0.473*** (0.187)
Fixed Effects:									
Cohort-Firm	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cohort-SIC-Year	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cohort-SIC-Size-Year	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cohort-SIC-Size-Custom-Year	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	18,381	16,039	11,374	18,266	15,960	11,301	25,301	23,225	19,029

Table 3

Summary statistics for China exporters by size quartiles. This table presents summary statistics for balance sheet characteristics of firms that export to China, broken down by size quartiles and treatment status (whether or not they were treated, namely suppliers of Chinese entities included in the BIS lists). SD refers to the standard deviation. Cash Flow equals operating income before depreciation minus interest and taxes, divided by lagged assets. Revenues is the logarithm of the total revenues (in millions), Sale is the logarithm of the total sales (in millions), ROA equals earnings before extraordinary items divided by lagged assets, CapEx is capital expenditures divided by lagged assets, Income equals operating income before depreciation divided by lagged assets, EBIT is earnings before interest and taxes divided by lagged assets, Interest equals interest expense divided by lagged assets, and Employees equals the logarithm of the total number of employees.

No. Obs.	Stat.	Size Q1			Size Q2			Size Q3			Size Q4		
		Tot.	Treated	Control	Tot.	Control	Treated	Tot.	Control	Treated	Tot.	Control	
	Full Sample	5,119	31	240	141	829	196	1,418	373	1,891			
Assets, \$m	Mean	11,946	1,057	145	2,589	711	1,404	1,189	27,778	25,280			
	Median	1,019	119	21	170	111	767	496	6,465	4,497			
	SD	48,333	1,750	462	9,783	3,501	1,670	1,803	54,993	72,858			
Cash Flow	Mean	0.015	-0.015	-0.286	-0.005	-0.068	0.086	0.001	0.125	0.072			
	Median	0.074	0.07	-0.08	0.032	0.016	0.088	0.064	0.115	0.088			
	SD	0.279	0.294	0.663	0.181	0.37	0.076	0.24	0.071	0.158			
Revenues	Mean	6.533	5.474	3.17	5.005	4.702	6.42	5.981	8.417	7.914			
	Median	6.099	4.8	2.884	5	4.44	6.375	6.02	8.26	8.001			
	SD	2.221	1.714	1.741	2.067	1.728	1.178	1.652	1.298	1.706			
Sales	Mean	6.523	5.474	3.17	5.004	4.702	6.42	5.983	8.418	7.898			
	Median	6.096	4.8	2.884	5	4.44	6.375	6.021	8.27	7.986			
	SD	2.218	1.714	1.741	2.064	1.728	1.178	1.655	1.298	1.703			
ROA	Mean	-0.041	-0.085	-0.324	-0.046	-0.134	0.024	-0.06	0.074	0.022			
	Median	0.027	0.011	-0.125	0	-0.036	0.036	0.011	0.069	0.045			
	SD	0.274	0.373	0.616	0.191	0.369	0.087	0.238	0.074	0.157			
CapEx	Mean	0.034	0.086	0.031	0.035	0.038	0.037	0.038	0.035	0.029			
	Median	0.023	0.026	0.018	0.024	0.025	0.026	0.027	0.023	0.021			
	SD	0.042	0.198	0.049	0.039	0.048	0.044	0.042	0.035	0.029			
Income	Mean	0.036	0.005	-0.267	0.009	-0.055	0.106	0.023	0.134	0.098			
	Median	0.098	0.073	-0.066	0.052	0.003	0.108	0.088	0.145	0.117			
	SD	0.263	0.285	0.611	0.18	0.33	0.079	0.227	0.074	0.157			
EBIT	Mean	-0.001	-0.057	-0.3	-0.025	-0.091	0.06	-0.016	0.111	0.064			
	Median	0.058	0.02	-0.106	0.008	-0.014	0.065	0.044	0.102	0.079			
	SD	0.272	0.322	0.614	0.19	0.36	0.081	0.235	0.07	0.162			
Interest	Mean	0.014	0.012	0.023	0.01	0.016	0.009	0.014	0.01	0.014			
	Median	0.008	0.009	0.005	0.004	0.005	0.008	0.007	0.01	0.009			
	SD	0.028	0.017	0.073	0.018	0.035	0.009	0.027	0.007	0.016			
Employees	Mean	7.727	6.778	4.58	6.25	6.062	7.656	7.23	9.502	8.968			
	Median	7.864	6.057	4.382	6.186	6.186	7.647	7.127	9.582	8.975			
	SD	2.055	1.702	1.632	2.09	1.675	1.373	1.447	1.398	1.558			

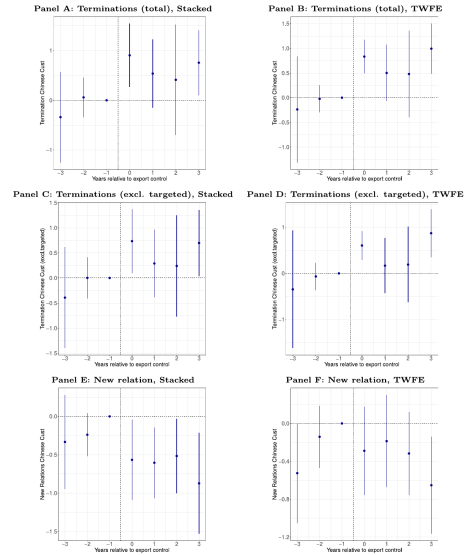


Fig. 2. Decoupling from Chinese customers. This figure shows the dynamic effects of export controls on the number of terminated Chinese customers in affected suppliers. Panels A, C and E show the coefficient plots for the number of terminated Chinese customers using the Poisson Pseudo Maximum likelihood (PPML) regression on the stacked regression while Panels B, D and F display the results on the total terminations with Chinese customers. Panels C and D show terminations with Chinese customers, excluding the targeted ones. Panels E and F display the results on the new relation with Chinese customers. Regressions include firm and industry-size quartile-lagged customer number quartile-year fixed effects. In the stacked regressions, the fixed effects are further interacted with the cohort indicator variable. The blue bars indicate 95% confidence intervals around the estimated dynamic coefficient (blue dot). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.6 Figure 2: dynamic decoupling from Chinese customers

Read: The dynamic estimates show that terminations rise after treatment and new Chinese relations fall after treatment. Pre-treatment coefficients are not the main driver of the result.

Economic message. The timing supports a policy-shock interpretation: export controls arrive first, and broad decoupling follows.

5.7 Table 5: number of customers after export controls

Read: Total customers decline after export controls, but domestic customers do not significantly increase.

Economic message. U.S. suppliers do not offset lost Chinese customers by quickly finding new U.S. customers. The customer base shrinks.

Table 5

Supply chain reconfiguration—Number of customers. This table presents the Poisson Pseudo Maximum Likelihood (PPML) regression results of the effect of export controls on supply chain configurations. Total Cust is the total number of customers. Domestic Cust is the number of domestic customers. Affected equals one for firms that within the previous year had a customer included in the BIS lists (Entity List, UVI, and MEU list) and Post equals one after the inclusion of such customer in the BIS lists. For each cohort, the control group includes never treated and not yet treated firms. SIC refers to the 2-digit standard industrial classification (SIC) code, and Size to the industry-specific size quartile of each firm. We require all firms to be exporting to China in the pre-treatment period. We double cluster the standard errors at the firm and year level. *** p<0.01, ** p<0.05, * p<0.1.

Dependent variables:	Total cust (1)	(2)	Domestic cust (3)	(4)
Affected×Post	−0.145** (0.064)	−0.138** (0.070)	−0.117 (0.075)	−0.097 (0.084)
Fixed Effects:				
Cohort-Firm	✓	✓	✓	✓
Cohort-SIC-Year	✓		✓	
Cohort-SIC-Size-Year		✓		✓
Observations	32,301	32,166	31,810	31,646

Table 6

Supply chain reconfigurations—Customer share. This table presents the Poisson Pseudo Maximum Likelihood (PPML) regression results of the effect of export controls on supply chain reconfigurations. Domestic Share is the ratio of the total number of domestic U.S. customers to the contemporaneous number of total customers. China Share is the ratio of the total number of Chinese customers to the contemporaneous number of total customers. Asia Share is the ratio of the total number of customers from Asia, excluding China, to the contemporaneous number of total customers. Asia Friend Share is the ratio of the total number of customers from South Korea, Japan, Taiwan, and Australia to the contemporaneous number of total customers. EU Share is the ratio of the total number of customers from European Union countries to the contemporaneous number of total customers. Affected equals one for firms that within the previous year had a customer included in the BIS lists (Entity List, UVI, and MEU list), and Post equals one after the inclusion of such customer in the BIS lists. For each cohort, the control group includes never treated and not yet treated firms. SIC refers to the 2-digit standard industrial classification (SIC) code. Size refers to the industry-specific size quartile of each firm. Custom refers to the classification of each firm into quartiles based on treatment firms' lagged total number of customers in each region. We require firms to export to China in the pre-treatment period. We double-cluster the standard errors at the firm and year level. *** p<0.01, ** p<0.05, * p<0.1.

Panel A: Domestic Share and China Share						
	Domestic Share (1)	(2)	China Share (3)	(4)		
Affected×Post	0.084*** (0.029)	0.092*** (0.032)	−0.336*** (0.075)	−0.390*** (0.120)		
Fixed Effects:						
Cohort-Firm	✓	✓	✓	✓		
Cohort-SIC-Size-Year						
Cohort-SIC-Size-Custom-Year				✓		
Observations	31,450	31,366	27,902	27,275		
Panel B: Other Customer Share						
	Asia Share (1)	(2)	Asia Friend Share (3)	(4)	EU Share (5)	(6)
Affected×Post	−0.026 (0.044)	−0.003 (0.040)	−0.042 (0.048)	0.003 (0.046)	−0.080 (0.058)	−0.002 (0.048)
Fixed Effects:						
Cohort-Firm	✓	✓	✓	✓	✓	✓
Cohort-SIC-Size-Year						
Cohort-SIC-Size-Custom-Year		✓				✓
Observations	29,027	28,823	28,207	27,855	27,751	27,313

5.8 Table 6: customer shares by region

Read: The domestic-customer share rises and the China share falls. However, customer shares from Asia, allied Asian countries, and Europe do not significantly rise.

Economic message. The higher domestic share mostly reflects fewer Chinese customers, not successful reshoring. There is little short-run evidence of friend-shoring to allied regions.

5.9 Figure 3: U.S. supplier stock-market reaction

Read: Cumulative abnormal returns for affected U.S. suppliers decline after BIS announcements in both the 3-factor and 5-factor models.

Economic message. Equity markets interpret export controls as bad news for affected U.S. suppliers, consistent with a loss of valuable customer relationships and limited replacement options.

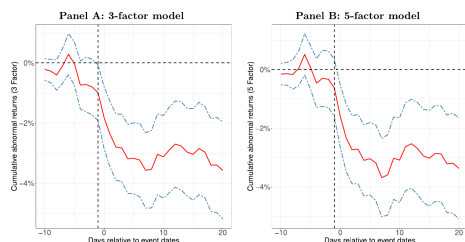


Fig. 3. Cumulative abnormal returns around announcement dates. This figure shows the cumulative abnormal returns (CAR) of affected suppliers in a [-10, 20] day window around the announcement date of the inclusion of a target entity in the BIS lists. Panel A shows CARs using the Fama-French 3-factor model (Fama and French, 1993) while Panel B uses the Fama-French 5-factor model (Fama and French, 2013). The dashed vertical line represents the day before announcement date. The solid red line represents the average CARs and the dot-dash blue lines represent the 95% confidence intervals. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 7

Real effects of export controls. This table presents the stacked regression results of the effect of export controls on cash flow, revenue, EBIT, capital expenditure and employment. Cash Flow equals operating income before depreciation minus interest and taxes, divided by lagged total assets. Revenues is the logarithm of the total revenues (in millions), EBIT is earnings before interest and taxes divided by lagged assets, CapEx is capital expenditures divided by lagged assets, Employees is the logarithm of the number of employees, and Affected equals one for firms that within the previous year had a customer included in the BIS lists (Entity List, UFL, and MEU list). Post equals one after the inclusion of such customer in the BIS lists. For each cohort, the control group includes never treated and not yet treated firms. SIC refers to the 2-digit standard industrial classification (SIC) code, Size refers to the industry-specific size quartile of each firm. We require all firms to be exporting to China in the pre-treatment period. We double cluster the standard errors at the firm and year level. *** p<0.01, ** p<0.05, * p<0.1.

Panel A: Full Sample					
Dependent variables:	Cash Flow	Revenues	EBIT	CapEx	Employees
	(1)	(2)	(3)	(4)	(5)
Affected $\times$ Post	-0.018** (0.007)	-0.093** (0.032)	-0.018* (0.008)	0.004 (0.003)	-0.076** (0.031)
Fixed Effects:					
Cohort-Firm	✓	✓	✓	✓	✓
Cohort-SIC-Size-Year	✓	✓	✓	✓	✓
Observations	31,613	31,605	31,615	31,570	31,459

Panel B: Restrictive Sample					
Dependent variables:	Cash Flow	Revenues	EBIT	CapEx	Employees
	(1)	(2)	(3)	(4)	(5)
Affected $\times$ Post	-0.017** (0.008)	-0.096*** (0.030)	-0.017* (0.008)	0.004 (0.003)	-0.076** (0.030)
Fixed Effects:					
Cohort-Firm	✓	✓	✓	✓	✓
Cohort-SIC-Size-Year	✓	✓	✓	✓	✓
Observations	26,778	26,784	26,780	26,744	26,608

5.10 Table 7: real effects on affected U.S. suppliers

Read: Affected suppliers experience lower cash flow, lower revenues, lower EBIT, and lower employment. Capital expenditures do not significantly decline.

Economic message. The valuation losses have real operating counterparts. Firms appear to adjust mainly through revenue, profitability, and labor rather than through an immediate cut in investment.

5.11 Table 8: bank lending to affected U.S. suppliers

Read: Total credit and term loans fall; spreads rise; maturities shorten. Credit lines and utilization do not show the same significant change.

Economic message. Export controls weaken the credit profile of affected suppliers. Banks respond by tightening loan quantity, price, and maturity terms.

Table 8

Bank lending to affected U.S. suppliers. This table presents the Poisson Pseudo Maximum Likelihood (PPML, columns 1–4) and OLS (columns 5–6) regression results of the effect of export controls on bank lending. Affected equals one for firms that within the previous year had a customer included in the BIS lists (Entity List, UFL, and MEU list) and Post equals one after the inclusion of such customer in the BIS lists. For each cohort, the control group includes never treated and not yet treated firms. SIC refers to the 2-digit standard industrial classification (SIC) code, Size refers to the industry-specific size quartile of each firm. We double cluster the standard errors at the firm and quarter level. *** p<0.01, ** p<0.05, * p<0.1.

Dependent variables:	Committed Total Credit	Committed Term Loans	Committed Credit Lines	Utilized Credit Lines	Spread	Maturity
	(1)	(2)	(3)	(4)	(5)	(6)
Affected $\times$ Post	-0.136* (0.073)	-0.630** (0.251)	-0.081 (0.068)	-0.197 (0.171)	0.179** (0.088)	-4.874*** (1.538)
Fixed Effects:						
Cohort-Firm	✓	✓	✓	✓	✓	✓
Cohort-SIC-Size-Quarter	✓	✓	✓	✓	✓	✓
Cohort-Bank-Quarter	✓	✓	✓	✓	✓	✓
Observations	356,012	356,012	356,012	356,012	174,368	202,016

Table 9

Decoupling from the U.S.—The Chinese perspective. This table presents the Poisson Pseudo Maximum Likelihood (PPML) regression results of the effect of export controls on Chinese firms' supply chain reconfigurations. Termination U.S. Supp is the total number of terminated relations with the U.S. suppliers. New Relations Chinese Supp is the number of new Chinese suppliers. New Relations U.S. Supp is the number of new U.S. suppliers. Targeted equals one for Chinese firms that are included in the BIS lists (Entity List, UFL, and MEU list) and Post equals one after the inclusion of such firms in the BIS lists. For each cohort, the control group includes never treated and not yet treated firms. Custom refers to the classification of each firm into quartiles based on the targeted firms' total number of customers prior to the treatment. We require all firms to be importing from U.S. suppliers in the pre-treatment period. We double cluster the standard errors at the firm and year level. *** p<0.01, ** p<0.05, * p<0.1.

Dependent variables:	Terminations U.S. Supp	New Relations Chinese Supp	New Relations U.S. Supp
	(1)	(2)	(3)
Targeted $\times$ Post	0.535* (0.291)	0.498* (0.283)	0.469*** (0.181)
Fixed Effects:			
Cohort-Firm	✓	✓	✓
Cohort-Year	✓	✓	✓
Cohort-Custom-Year	✓	✓	✓
Observations	156,913	156,004	183,178

5.12 Table 9: Chinese-target supply-chain response

Read: Targeted Chinese firms terminate U.S. supplier relations and form more new relationships with Chinese suppliers. New U.S. supplier formation falls but is not statistically significant.

Economic message. Chinese targets respond more proactively than U.S. suppliers: decoupling from the U.S. is accompanied by domestic reshoring.

5.13 Table 10: Chinese-target supplier shares

Read: The total number of suppliers does not significantly change, but the Chinese supplier share rises and the U.S. supplier share falls.

Economic message. Chinese firms substitute across supplier origins. They preserve supplier-network size while changing its composition away from the U.S. and toward China.

Table 10
Decoupling from the U.S.—Customer shares. This table presents the Poisson Pseudo Maximum Likelihood (PPML) regression results of the effect of export controls on Chinese firms' supply chain configurations. Total Suppliers is the total number of suppliers. China Supplier Share is the ratio of the total number of Chinese suppliers to the contemporaneous number of total suppliers. U.S. Supplier Share is the ratio of the total number of U.S. suppliers to the contemporaneous number of total suppliers. Targeted equals one for Chinese firms that are included in the BIS lists (Entity List, UVL, and MEU list) and Post equals one after the inclusion of such Chinese firms in the BIS lists. For each cohort, the control group includes never treated and not yet treated firms. Custom refers to the classification of each firm into quartiles based on the targeted firms' total number of customers prior to the treatment. We require all control firms to be importing from U.S. in the pre-treatment period. We double cluster the standard errors at the firm and year level. *** p<0.01, ** p<0.05, * p<0.1.

Dependent Variables:	Total Suppliers (1)	(2)	China Supplier Share (3)	(4)	U.S. Supplier Share (5)	(6)
Targeted xPost	0.066 (0.124)	0.115 (0.111)	0.322*** (0.120)	0.318*** (0.119)	-0.373*** (0.134)	-0.328** (0.135)
Fixed Effects:						
Cohort-Firm	✓	✓	✓	✓	✓	✓
Cohort-Year	✓	✓	✓	✓	✓	✓
Cohort-Custom-Year		✓		✓		✓
Observations	239,597	239,597	172,889	172,888	182,475	182,473

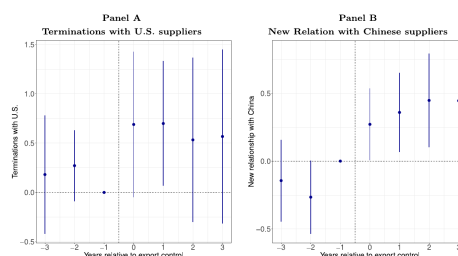


Fig. 4. Chinese firms' supply chain reconfiguration. This figure shows the dynamic effects of export controls on the supply chains of Chinese firms targeted by U.S. export controls. Panel A shows the coefficient plot for the terminations with U.S. suppliers using the Poisson Pseudo Maximum Likelihood (PPML) regression on the stacked regression. Panel B displays the dynamic effect on new relations with Chinese suppliers using the Poisson Pseudo Maximum Likelihood (PPML) regression on the stacked regression. Regressions include cohort-firm and cohort-customer number quartile-year fixed effects. The blue bars indicate 95% confidence intervals around the estimated dynamic coefficient (blue dot). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.14 Figure 4: dynamic Chinese supply-chain reconfiguration

Read: The event-study plots show post-treatment increases in terminations with U.S. suppliers and new relationships with Chinese suppliers.

Economic message. The Chinese-side reconfiguration is not simply a pre-existing trend. It emerges around the export-control event.

5.15 Table 11: supply-chain circumvention

Read: Non-U.S. suppliers to Chinese targets, especially those in allied regions, experience revenue increases after export controls. EBIT effects are positive but not statistically significant.

Economic message. Some demand lost by U.S. suppliers may shift to non-U.S. suppliers. This weakens the idea that export controls simply deny Chinese firms access to all similar technologies.

5.16 Figure 5: stock-market reaction of targeted Chinese firms

Read: Targeted Chinese firms experience large negative CARs, around -8.2% to -9.0% depending on the factor model.

Table 11

Supply chain circumvention. This table presents the regression results of the effect of export controls on the revenues and EBIT of suppliers from all regions (excluding the U.S.) and suppliers in allied regions (European Union, South Korea, Japan, Taiwan, Australia, and Canada). Revenues and EBIT are as defined in Table 7. Affected equals one for firms that within the previous year had a customer included in the BIS lists (Entity List, UVI, and MEU list) and Post equals one after the inclusion of such customer in the BIS lists. For each cohort, the control group includes never treated and not yet treated firms. SIC refers to the 2-digit standard industrial classification (SIC) code, and Size refers to the size quartile of each firm in each region (Europe, Asia, etc.). We require all firms to be exporting to China in the pre-treatment period. We double-cluster the standard errors at the firm and year level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Dependent variables:	Revenues		EBIT	
	(1)	(2)	(3)	(4)
Affected $\times$ Post	0.157*** (0.049)	0.127** (0.048)	0.156 (0.126)	0.168 (0.153)
Fixed Effects:				
Cohort-Firm	✓	✓	✓	✓
Cohort-SIC-Year	✓		✓	
Cohort-SIC-Size-Year		✓		✓
Observations	95,711	95,711	96,181	96,181

Panel B: Treated firms in all regions				
Dependent variables:	Revenues		EBIT	
	(1)	(2)	(3)	(4)
Affected $\times$ Post	0.043* (0.023)	0.024 (0.024)	0.036 (0.027)	0.039 (0.030)
Fixed Effects:				
Cohort-Firm	✓	✓	✓	✓
Cohort-SIC-Year	✓		✓	
Cohort-SIC-Size-Year		✓		✓
Observations	328,588	328,588	329,418	329,418

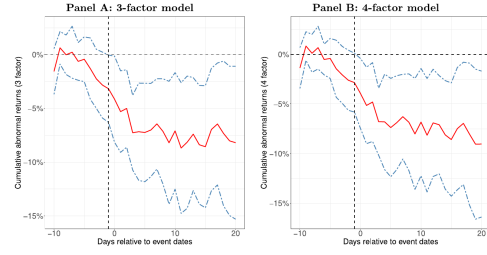


Fig. 5. Cumulative abnormal returns on targeted Chinese firms. This figure shows the cumulative abnormal returns (CAR) of targeted Chinese firms in a [-10, 20] day window around the announcement date of the inclusion of a target entry in the BIS lists. Panels A and B show CARs using the 3- and 4-factor models of [Liu et al. \(2019\)](#), respectively. The dashed vertical line represents the day before the announcement date. The solid red line represents the average CARs and the dot-dash blue lines represent the 95% confidence intervals. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Economic message. Chinese targets also bear costs. But the paper emphasizes that aggregate losses on the U.S. supplier side are larger because each Chinese target is connected to multiple U.S. suppliers and many affected U.S. suppliers are large.

6 Short summary

- Export controls force immediate decoupling between U.S. suppliers and targeted Chinese customers.
- The decoupling spills beyond the directly targeted Chinese firms: affected U.S. suppliers also cut non-target Chinese links and create fewer new Chinese links.
- Affected U.S. suppliers do not quickly replace lost Chinese demand with domestic, allied-Asia, or European customers.
- U.S. suppliers experience negative stock-market reactions, lower revenue, lower profitability, lower employment, and tighter bank credit.
- Chinese targets respond by reshoring supplier relationships toward China and by shifting demand to non-U.S. suppliers.
- Chinese targets also lose market value, but the aggregate market-capitalization loss is larger for U.S. suppliers.

7 Conclusion

7.1 Core conclusion

Export controls achieve the direct goal of cutting off targeted Chinese firms from U.S. suppliers. However, the domestic cost is large. Affected U.S. suppliers lose customers, cannot easily replace them, suffer valuation losses, experience real operating declines, and face tighter bank credit.

7.2 Economic intuition

Supply-chain relationships are sticky and relationship-specific. Losing a major Chinese customer is not equivalent to selling the same product immediately to a domestic or allied buyer. If the customer relationship contains match-specific knowledge, regulatory certification, production adaptation, or strategic complementarity, then forced decoupling destroys valuable business relationships.

7.3 Policy takeaway

The paper does not claim that export controls have no national-security value. Instead, it shows that the cost side is substantial and falls heavily on the domestic firms producing strategic technologies. If export controls remain a central geoeconomic tool, they may need complementary policies that stimulate domestic or allied demand for the same technologies.